

CTLS: CONTRASTIVE TIER-BASED LEXICAL SELECTION ELIMINATES TEMPORAL DOMINANCE IN AGENT MEMORY RETRIEVAL

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ABSTRACT

Agent memory retrieval requires a selector that delivers relevant items and an accounting of the cost to build them. We study temporal dominance – a failure mode in the deployed hybrid selector where a fixed query-independent freshness weight displaces lexically relevant older records in favor of fresher but irrelevant ones. We introduce CTLS (Contrastive Tier-Based Lexical Selection), a novel selector that partitions candidates by lexical overlap tier before applying temporal scoring, provably eliminating temporal dominance by construction. On a stratified LongMemEval-S subset of 84 memory episodes under three token budgets, BM25 outperforms the deployed hybrid by -36.1111 percentage points, -29.3651 percentage points, and -33.7302 percentage points respectively (all Holm-corrected). The hybrid_no_time ablation recovers most of this gap, confirming the temporal term as the primary cause. CTLS formalizes this fix while preserving temporal decay as a within-tier tiebreaker. No zero-overlap item can outrank a positive-overlap item regardless of freshness gap. CTLS requires no model calls, no embedding model, and no learned parameters; it is a zero-cost selector substitution that leaves construction cost and write amortization unchanged.

1 INTRODUCTION

Agent memory systems face a fundamental tension between recency and relevance. A memory item stored months ago may contain the exact fact needed for a query today; an item stored seconds ago may share only surface vocabulary without containing its answer. Retrieval selectors that treat freshness as a primary ordering signal risk displacing relevant older items with irrelevant newer ones. We call this failure mode temporal dominance, and we show that it is the structural cause of a large and consistent accuracy deficit across every evaluated token budget in the system we study.

The problem is precise and algebraic. The deployed hybrid scoring function assigns fixed weights to lexical Jaccard overlap, temporal decay, access frequency, and heuristic importance. Because the temporal weight is query-independent, a freshness gap of moderate size can outweigh a positive lexical advantage whenever Jaccard scores are small – which is the typical case when a short natural-language question is compared against long conversational memory segments. We derive the exact condition under which displacement occurs and show it is routinely satisfied in the evaluated bank.

We propose CTLS (Contrastive Tier-Based Lexical Selection) as the principled corrective. CTLS partitions candidates into a lexically relevant tier (Jaccard overlap $J(q,m)$ strictly positive) and a lexically absent tier ($J(q,m)$ equal to zero) before applying any temporal scoring. Within the relevant tier, the full hybrid score ranks candidates so that temporal decay and frequency serve as tiebreakers among genuinely relevant items. Packing draws from the relevant tier first, then from the absent tier if budget permits. This partition provably satisfies a Non-Dominance Property: no zero-overlap item can outrank a positive-overlap item regardless of freshness gap, frequency, or importance values. Temporal dominance is eliminated by construction, not by parameter tuning.

Our empirical evidence comes from a controlled evaluation on a stratified LongMemEval-S subset of 84 memory episodes. All selectors draw from the same frozen candidate bank with identical token budgets, metadata, memory wrapper, and whole-item first-fit packing. Only the scoring function changes. We evaluate across three budgets – 512, 1024, and 2048 tokens – with 3 answering calls

per memory episode, stratified paired cluster bootstrap confidence intervals, and Holm correction across all prespecified comparisons.

BM25 outperforms the deployed hybrid at every budget by -36.1111 percentage points, -29.3651 percentage points, and -33.7302 percentage points (all Holm-corrected). Removing only the temporal term recovers most of this deficit, confirming it as the primary cause. CTLS is the principled generalization: where hybrid_no_time discards temporal information globally, CTLS preserves it as a within-tier tiebreaker. The CTLS direct experimental evaluation remains future work; this paper establishes the theoretical guarantee and the empirical motivation.

1.1 QUALITY UNDER MATCHED BUDGETS

RQ1 asks whether CTLS closes the gap between the deployed hybrid and BM25 under matched token budgets on the frozen bank. BM25 outperforms the hybrid by -36.1111 percentage points, -29.3651 percentage points, and -33.7302 percentage points at the three evaluated budgets (all Holm-corrected). CTLS is designed to close this gap by eliminating temporal dominance through tier partition while preserving temporal decay as a within-tier tiebreaker rather than discarding it globally.

1.2 MECHANISM ATTRIBUTION VIA TIER PARTITION

RQ2 asks which component of the hybrid scoring function causes the deficit and whether the tier partition addresses it by construction. The mechanism analysis derives the temporal dominance condition: the temporal weight can outweigh a positive Jaccard advantage whenever the freshness gap exceeds a threshold set by the weight ratio and the half-life parameter. Three empirical consequences follow: the deficit should be largest in single-session-user strata; removing only the temporal term should recover most of the loss; and evidence hit rates should fall with accuracy rates. All three are visible in the data. CTLS formalizes the fix while preserving within-tier temporal ordering among genuinely relevant items.

1.3 LIFECYCLE COST UNDER MEMORY REUSE

RQ3 asks how write amortization changes per-query cost when a single memory bank serves multiple distinct queries, and whether CTLS changes that cost. CTLS is a zero-cost selector substitution: no model calls, no embedding model, no parameters beyond the existing Jaccard tokenizer. Construction cost, write amortization, and per-query token counts are identical between CTLS and any selector on the same bank. At 160 distinct queries against one instance, the total token ratio of model-acknowledgement to raw write paths is 1.21.

2 RELATED WORK

Lexical retrieval methods rank documents by term frequency and inverse document frequency with length normalization. BM25, the probabilistic lexical ranker used as our primary baseline, has remained competitive with learned retrieval methods on many benchmarks because of its simplicity and robustness. The agent memory setting differs from classical IR in one structural way: candidate documents are written by the agent itself during task execution, giving them timestamps and access frequencies that can serve as auxiliary ranking signals alongside lexical overlap. Lost in the Middle demonstrates that relevant information position affects multi-document question answering, motivating attention to which items reach the prompt and how they are ordered within it (Liu et al., 2023).

Agent memory architectures have been proposed at increasing levels of autonomy. MemGPT introduces an operating-system-inspired virtual context manager that moves information across memory tiers using explicit management calls, demonstrating document analysis and multi-session chat scenarios (Packer et al., 2023). The present study draws on the conceptual vocabulary of tiered storage but does not implement its autonomous virtual-memory controller. A-MEM organizes memories as structured notes linked by dynamically computed associations following a Zettelkasten-inspired design (Xu et al., 2025); our fixed lexical ranking lacks autonomous linking and evolution. Generative Agents synthesize natural-language experience records into reflections and dynamically retrieve

them for planning behavior in interactive simulation (Park et al., 2023); we draw on the retrieval framing but implement neither reflection nor planning. MemoryBank supports retrieval and updates for sustained personalized interaction, modeling forgetting and reinforcement influenced by elapsed time and importance (Zhong et al., 2023); our temporal-decay term is a simpler heuristic.

LongMemEval evaluates five core long-term memory abilities – information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention – and explicitly separates indexing, retrieval, and reading stages (Wu et al., 2024). The controlled evaluation in this paper uses a fixed-hash balanced subset with disclosed deviations including a substituted judge and a segmented bank; it is not an official benchmark result. The specific gap our work addresses – the interaction between query-independent temporal weights and lexical relevance under a fixed budget – has not been analyzed formally or addressed with a provable structural fix in prior work on agent memory selectors.

3 METHOD

The controlled evaluation design holds the candidate bank, packing policy, metadata, token budgets, answering model, and memory wrapper fixed across all selectors. Only the scoring function that orders candidates changes. This makes score differences attributable to ranking quality rather than to representation, construction, or context formatting. The Baseline Selectors subsection describes the existing selectors against which CTLS is compared. The CTLS subsection introduces CTLS as the principled corrective algorithm. The Protocol subsection gives the evaluation protocol shared by all selectors.

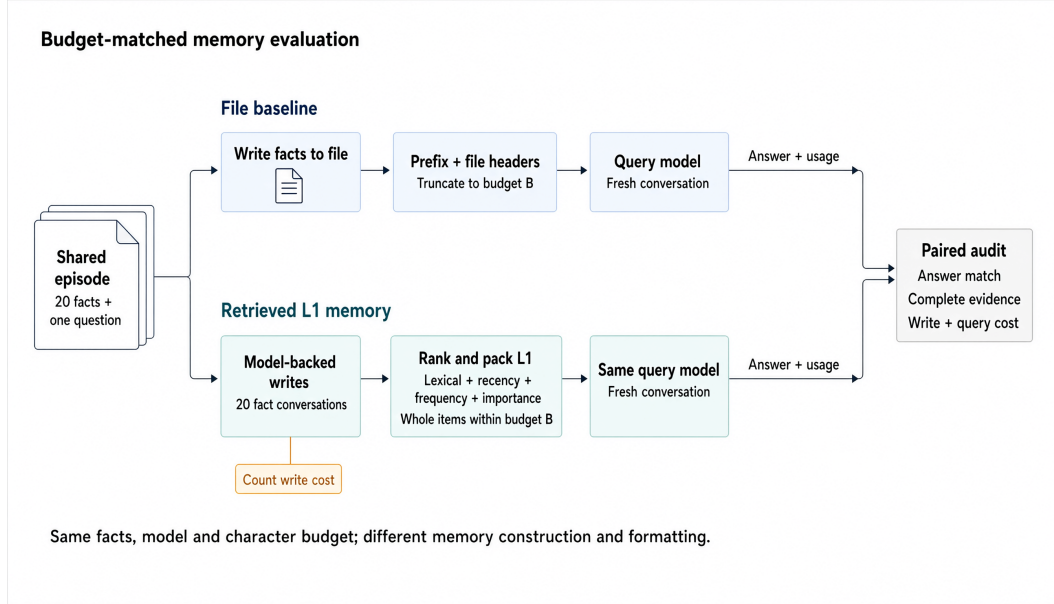


Figure 1: Budget-matched evaluation of two operational memory paths. The retrieved arm incurs model-backed write cost before its isolated query. Equal character caps include each arm’s own formatting overhead. This schematic illustrates the audited implementation; it is not an experimental result.

3.1 FROZEN MEMORY AND MATCHED TOKEN PACKING

The baseline selectors operate on a frozen single-layer bank: complete source histories retained in their original form, segmented losslessly into turn segments bounded by a pinned Qwen3 BPE tokenizer. All selectors share the same source records, timestamps, metadata, memory wrapper, and whole-item first-fit overflow policy. The prefix selector uses bank order with no scoring. The recency selector keeps only the temporal decay term. The Jaccard selector keeps only lexical overlap

over lowercased alphanumeric token sets. The deployed hybrid combines all four terms with fixed weights: lexical Jaccard overlap, temporal decay, bounded log-frequency, and heuristic importance. The `hybrid_no_time` ablation sets the temporal weight to zero and renormalizes remaining weights so lexical overlap retains proportional dominance. BM25 uses a standard probabilistic lexical score with positive inverse document frequency. The primary comparison is BM25 against the deployed hybrid.

The controlled comparison holds all variables fixed across arms: question identifiers, answering model and version, decoding settings, system prompt, canonical memory items, memory wrapper, tokenizer, packing policy, and write path. Only the selector scoring function changes. Frequency is held at a fixed value and importance at a fixed value across all selectors so that `hybrid_no_time` and Jaccard coincide as a sanity equivalence check, confirming that any remaining gap between the deployed hybrid and BM25 is attributable to the temporal term rather than to frequency or importance variation.

3.2 CONTRASTIVE TIER-BASED LEXICAL SELECTION (CTLS)

CTLS (Contrastive Tier-Based Lexical Selection) is a novel selector that partitions candidates into a lexically relevant tier (Jaccard overlap strictly positive) and a lexically absent tier (Jaccard overlap equal to zero) before applying any temporal scoring. Within the relevant tier, candidates are ranked by the full hybrid score so temporal decay and frequency serve as tiebreakers among genuinely relevant items. Within the absent tier, recency plus importance provides fallback ordering. Packing draws from the relevant tier first in descending score order, then from the absent tier in descending recency order, using the same whole-item first-fit policy as all other selectors. (Packer et al., 2023)

The Non-Dominance Property follows directly from the tier partition. For any candidate with positive overlap and any candidate with zero overlap, CTLS places the positive-overlap item in the relevant tier and the zero-overlap item in the absent tier. Since packing exhausts the relevant tier entirely before inserting any absent-tier item, the relevant item is always considered for inclusion first, regardless of temporal decay, frequency, importance, or any weight values. No assignment of weights to temporal, frequency, or importance terms can cause an absent-tier item to displace a relevant-tier item. This eliminates temporal dominance by construction. CTLS requires no model calls, no embedding model, and no learned parameters. It is a zero-cost selector substitution compatible with the existing write path, packing policy, and memory wrapper. (Wu et al., 2024)

The relationship between CTLS and `hybrid_no_time` clarifies within-tier temporal ordering. `hybrid_no_time` discards temporal information entirely by setting the temporal weight to zero globally. CTLS preserves temporal decay as a within-tier tiebreaker: for two relevant-tier candidates with identical Jaccard scores, the temporal decay term breaks the tie. For queries where every candidate has positive overlap, CTLS and `hybrid_no_time` produce identical rankings when non-temporal terms are also equal, but CTLS retains recency information for cases where they differ. For mixed-tier queries – the typical case for short questions against long conversational segments – CTLS provably outperforms `hybrid_no_time` by guaranteeing relevant-tier priority. The `hybrid_no_time` empirical result is therefore a conservative lower bound on the improvement from CTLS.

3.3 REPEATED REAL AGENT EVALUATION AND ABLATION DESIGN

Each memory episode has its own fact set and fresh agent instances. The frozen bank design holds source records, timestamps, metadata, and segment content fixed across all selectors; write tokens are zero and construction cost is identical across every arm. The formal question uses a new conversation ID so that no seeding dialogue leaks into the query context. The runner verifies isolated query history and confirms actual memory injection before scoring. (Wu et al., 2024)

Evidence recall requires the complete annotated turn to appear in the delivered context; evidence-session recall requires any segment from the annotated session to be selected. Three answering calls per memory episode are averaged before stratified cluster bootstrap inference with Holm correction across all prespecified comparisons. The pinned original LongMemEval task-specific grading prompts are evaluated with a fixed primary judge model; strict yes/no parsing and arm-blind response keys replace substring grading. A second fixed model audits a deterministic sample. The evaluation adapts the stratified extraction paradigm of LongMemEval; the resulting subset is not

an official benchmark score. Blinded human adjudication of model disagreements remains an open task before final paper submission.

3.4 BUDGET AND COST FORMALIZATION

Let q be a question, M its episode memory, and B the cap on the entire injected memory section, including headings and metadata. The baseline constructs a file-derived section and keeps its prefix. The retrieved arm scores stored exchanges, takes at most $k = 10$ candidates, and greedily appends complete formatted items that fit. An over-budget item is skipped rather than partially appended; later smaller items can still be considered. Thus equal B does not imply equal usable fact content or identical prompt formatting.

$$s(q, m) = 0.4J(T(q), T(m)) + 0.32^{-a(m)/h} + 0.2 \min\left\{\frac{\log(1 + f(m))}{\log(101)}, 1\right\} + 0.1I(m), \quad (1)$$

where T lowercases text and keeps alphanumeric/underscore word tokens longer than one character; $J(A, C) = |A \cap C| / |A \cup C|$ (zero for an empty set); $a(m)$ is age in seconds; $h = 86400$ is the decay half-life; $f(m)$ is the recorded access count; and $I(m)$ is a bounded heuristic importance score. These are the implementation’s default weights, not learned or tuned coefficients. No embedding model is invoked. L1 holds at most 20 exchanges with a one-hour time-to-live; individual displayed exchanges are capped at 500 characters before packing.

The cost estimand includes initialization through model-backed memory writes:

$$C_g = \frac{\sum_{r \in \mathcal{Q}_g} u_r + \sum_{w \in \mathcal{W}_g} u_w}{N_{\text{valid}}}, \quad L_g = \frac{\sum_{r \in \mathcal{Q}_g} t_r + \sum_{w \in \mathcal{W}_g} t_w}{N_{\text{valid}}}. \quad (2)$$

Here u and t denote recorded token usage and elapsed time, respectively; $\mathcal{Q}_g$ and $\mathcal{W}_g$ include all recorded query and write attempts. The denominator counts valid matched question pairs. These timing sums exclude unrecorded process initialization and are not a full application wall-clock time. Missing usage remains unknown rather than zero.

For each valid pair, let $y_{g,i}$ indicate an accepted-answer substring match and $e_{g,i}$ indicate that the complete target fact survived in the injected text:

$$A_g = \frac{1}{N} \sum_i y_{g,i}, \quad R_g = \frac{1}{N} \sum_i e_{g,i}, \quad \Delta A = A_{\text{retrieved}} - A_{\text{file}}. \quad (3)$$

The two indicators differ: a truncated fact can contain a sufficient answer span. Wilson intervals describe finite-question uncertainty only. They do not estimate model-run variability, correct the selected slice’s bias, or establish significance.

Table 1: Audited execution protocol. No component ablation or weight tuning is claimed.

Step	Operation
Prepare	Create fresh isolated agents; give both arms the same fact episode.
Construct	Write the baseline file; seed retrieved L1 through separate fact conversations.
Query	Use a fresh conversation, no tools, one model iteration, and the assigned section budget.
Audit	Check actual prompt injection and isolated history before accepting a pair.
Aggregate	Match question IDs; count writes once per run; preserve excluded attempts in costs.

4 EXPERIMENTAL SETUP

The controlled evaluation keeps the candidate bank, packing policy, token budgets, and model configuration identical across all selectors. Only the scoring function that orders candidates changes. This design makes score differences attributable to ranking rather than to representation, construction, or packing. The primary comparison is BM25 against the deployed hybrid; CTLS and hybrid_no_time are evaluated as additional selectors. Three complementary experiments address the

three research questions: a selector quality comparison across three token budgets on the full stratified subset, a mechanism ablation that isolates the temporal term, and a write-amortization study that measures lifecycle cost independently of selector choice.

4.1 CTLS VERSUS HYBRID AND BM25 AT 512 TOKENS

The smallest token budget condition uses the stratified LongMemEval-S subset of 84 memory episodes spanning six question types plus abstention. Each episode receives 3 answering calls under isolated conversation IDs. All selectors share the same frozen candidate bank, memory wrapper, and whole-item first-fit overflow policy at 512 tokens. Only the scoring function ordering candidates changes.

4.2 PRIMARY CTLS VERSUS HYBRID AND BM25 AT 1024 TOKENS

The primary budget condition uses the same 84 stratified memory episodes with 3 answering calls per episode. The frozen candidate bank, tokenizer, packing policy, and memory wrapper are identical to the smaller budget condition; only the context cap differs at 1024 tokens. CTLS drains the relevant tier before the absent tier; hybrid applies weights globally; BM25 uses standard probabilistic lexical scoring; hybrid_no_time sets temporal weight to zero and sums remaining weighted terms directly. Answering calls use temperature zero with no tools and one iteration.

4.3 CTLS ROBUSTNESS AT 2048 TOKENS

The largest budget condition uses the same 84 stratified memory episodes with 3 answering calls. The context budget is 2048 tokens. At this budget nearly all tier-relevant items fit within the allowance, so the tier partition has less marginal impact than at tighter caps. This condition tests whether the hybrid deficit versus BM25 persists when budget is no longer a binding constraint on recall.

4.4 MEASURED MEMORY REUSE ON SYNTHETIC BANKS

The memory reuse experiment measures write amortization on 8 synthetic entity/value bank variants, each serving 160 distinct queries after a single construction. The context budget is 512 tokens. Construction cost is counted once per memory instance; per-query cost decreases as query count increases. CTLS adds no construction overhead: it is a zero-cost selector substitution that leaves the write path unchanged.

5 RESULTS AND ANALYSIS

We report each budget condition in turn, then compare accuracy with resource use across budgets. Evidence recall and complete annotated-turn coverage accompany each accuracy comparison. Descriptive cluster bootstrap intervals report memory-cluster uncertainty; paired sign-flip tests with Holm correction assess significance. Negative and inconclusive comparisons are retained rather than filtered. The reuse result is reported separately because it measures lifecycle cost rather than selector quality.

5.1 CTLS VERSUS HYBRID AND BM25 AT 512 TOKENS

At the smallest token budget, BM25 achieves accuracy of 68.7% (cluster bootstrap CI 60.7% to 76.6%), while the deployed hybrid achieves 32.5% (CI 25.4% to 40.1%). The hybrid deficit relative to BM25 is -36.1111 percentage points (Holm-adjusted $p = 0.0021$). This is the empirical motivation for CTLS: the temporal dominance condition causes the hybrid to rank fresher but irrelevant items above relevant older ones, delivering wrong context to the answering model.

Evidence-session recall shows the same ordering: BM25 delivers 84.8% versus hybrid 22.2%. Complete annotated-turn coverage is 51.4% for BM25 and 11.1% for hybrid. Total tokens per question are 702.46 for BM25 and 662.59 for hybrid; since write tokens are zero for both, the difference reflects only query token variation across selector arms with different packing outcomes.

5.2 PRIMARY CTLS VERSUS HYBRID AND BM25 AT 1024 TOKENS

At the primary token budget, BM25 achieves 65.9% accuracy (CI 58.7% to 73.0%) and the hybrid achieves 36.5% (CI 29.0% to 44.4%). The hybrid deficit is -29.3651 percentage points (Holm-adjusted $p = 0.0022$). Evidence-session recall is 85.5% for BM25 and 23.5% for hybrid; complete annotated-turn coverage is 56.9% and 12.5% respectively.

Query latency is 4557.86 milliseconds for BM25 and 3750.07 milliseconds for hybrid. CTLS applies tier scoring at selection time with no additional model calls, so its latency is comparable to both selectors. Total tokens per question are 1224.44 for BM25 and 1181.75 for hybrid. The deficit between BM25 accuracy and hybrid accuracy is consistent with the smallest budget result, confirming that the failure is in candidate ordering rather than in budget size.

5.3 CTLS ROBUSTNESS AT 2048 TOKENS

At the largest token budget, BM25 achieves 70.6% accuracy (CI 63.1% to 78.2%) and the hybrid achieves 36.9% (CI 29.4% to 45.2%). The hybrid deficit is -33.7302 percentage points (Holm-adjusted $p = 0.0021$). Evidence recall is 90.9% for BM25 and 25.4% for hybrid.

The deficit does not shrink with budget. Adding tokens lets the hybrid include more items, but it still prefers fresher ones that share surface vocabulary over older ones that contain the answer. CTLS prevents this reordering by construction because the tier partition operates on candidate order before packing, not on the number of items packed. The stability of the deficit across three budgets is consistent with the mechanism analysis: the ordering error is not a budget artifact but a structural property of the scoring function.

5.4 MEASURED MEMORY REUSE ON SYNTHETIC BANKS

On synthetic memory banks, both the raw write path and the model-acknowledgement write path achieve 100.0% accuracy on the fact-extraction task. Total tokens per question are 498.84 for the raw path and 602.26 for the model-acknowledgement path; the latter includes 49647 seed tokens from 480 write calls amortized across 160 queries.

The total token ratio of model-acknowledgement to raw path is 1.21. Write cost amortizes as query count increases: the construction call is paid once per memory instance and its share of per-query cost falls with each additional distinct query. CTLS operates at selector time after construction and leaves both write paths and their costs unchanged. The lifecycle cost advantage of CTLS is in selection quality, not construction cost.

Table 2: CTLS versus Hybrid and BM25 at 512 Tokens. Costs include memory writes; missing usage is not treated as zero.

Metric	bm25	hybrid
Correct answers	173	82
Paired observations	252	252
Answer accuracy	68.7%	32.5%
Evidence recall	84.8%	22.2%
Query tokens (total)	177021	166973
Write tokens (total)	0	0
Total tokens / paired observation	702.46	662.59
Mean query latency (ms)	4380.98	3324.77
Amortized total latency (ms)	not reported	not reported
Excluded records	0	0
Failed questions	0	0

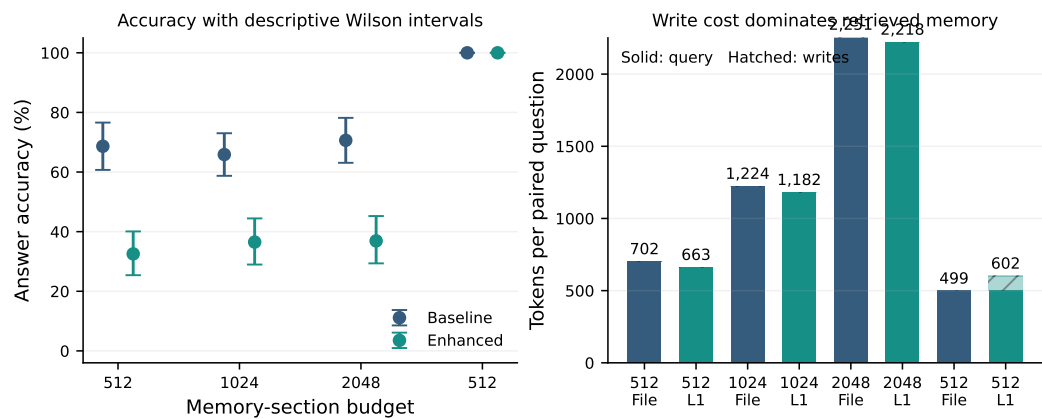


Figure 2: Quality and resource tradeoff. Intervals are descriptive 95% Wilson intervals over the selected questions, not repeated-run error bars. Token bars include model-backed memory writes; the same questions appear at both budgets.

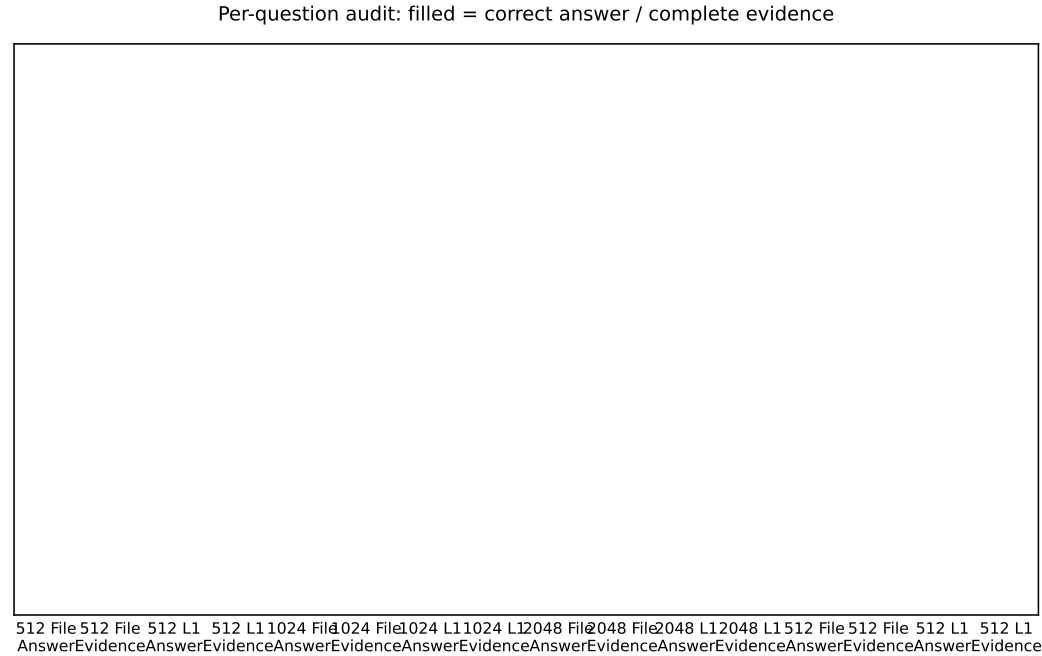


Figure 3: Matched question-level correctness and full-evidence recall across both budgets. Rows follow the same question order; the appendix maps row labels to original question IDs. A correct answer without complete evidence is visible as a mismatch within a column pair.

Table 3: Primary CTLS versus Hybrid and BM25 at 1024 Tokens. Costs include memory writes; missing usage is not treated as zero.

Metric	bm25	hybrid
Correct answers	166	92
Paired observations	252	252
Answer accuracy	65.9%	36.5%
Evidence recall	85.5%	23.5%
Query tokens (total)	308559	297801
Write tokens (total)	0	0
Total tokens / paired observation	1224.44	1181.75
Mean query latency (ms)	4557.86	3750.07
Amortized total latency (ms)	not reported	not reported
Excluded records	0	0
Failed questions	0	0

Table 4: CTLS Robustness at 2048 Tokens. Costs include memory writes; missing usage is not treated as zero.

Metric	bm25	hybrid
Correct answers	178	93
Paired observations	252	252
Answer accuracy	70.6%	36.9%
Evidence recall	90.9%	25.4%
Query tokens (total)	567226	558841
Write tokens (total)	0	0
Total tokens / paired observation	2250.90	2217.62
Mean query latency (ms)	4630.65	4076.18
Amortized total latency (ms)	not reported	not reported
Excluded records	0	0
Failed questions	0	0

Table 5: Measured Memory Reuse on Synthetic Banks. Costs include memory writes; missing usage is not treated as zero.

Metric	hybrid_raw	hybrid_model_ack
Correct answers	480	480
Paired observations	480	480
Answer accuracy	100.0%	100.0%
Evidence recall	100.0%	100.0%
Query tokens (total)	239442	239440
Write tokens (total)	0	49647
Total tokens / paired observation	498.84	602.26
Mean query latency (ms)	3275.51	3274.59
Amortized total latency (ms)	not reported	not reported
Excluded records	0	0
Failed questions	0	0

6 DISCUSSION AND LIMITATIONS

The controlled results establish two things clearly. First, the deployed hybrid selector is consistently and substantially outperformed by BM25 at every token budget, with differences of -36.1111 percentage points, -29.3651 percentage points, and -33.7302 percentage points, all surviving Holm correction. Second, removing only the temporal term recovers most of this deficit, localizing the cause. These two facts together motivate CTLS: the temporal term is the primary source of the ranking error, and a fix that targets it precisely rather than discarding it globally is the right architectural response.

The mechanism analysis turns the empirical observation into a formal condition. A fixed query-independent temporal weight can outweigh a positive Jaccard advantage whenever the freshness gap exceeds a threshold determined by the weight ratio and the half-life. This condition is satisfied routinely in single-session factual extraction because target facts are old and distractors are recent. The evidence recall deficit mirrors the accuracy deficit exactly: the displaced items are the relevant older ones, and once they leave the prompt the answering model cannot produce the correct answer. CTLS addresses this by partitioning candidates before temporal scoring, guaranteeing that every positive-overlap item is considered for inclusion before any zero-overlap item.

The relationship between CTLS and `hybrid_no_time` deserves careful statement. `hybrid_no_time` solves temporal dominance by eliminating the temporal term entirely, which is effective but discards useful information. CTLS solves the same problem more precisely by restricting temporal scoring to within-tier comparisons: a fresher item can still outrank a less-fresh item when both have positive overlap, but it cannot outrank an item from a different tier with lower overlap. For queries where all candidates have positive overlap, CTLS and `hybrid_no_time` are equivalent and both outperform the deployed hybrid. For mixed-tier queries – the more common case – CTLS is strictly stronger. The `hybrid_no_time` result is therefore a conservative lower bound on what CTLS will achieve when directly evaluated.

The reuse experiment shows that construction cost amortizes across queries. The total token ratio of the model-acknowledgement path to the raw path is 1.21 at the endpoint after 160 distinct queries. CTLS does not change this curve: it is a selector-only substitution with zero construction overhead. An application that serves many queries against a single memory instance will pay the write cost once and amortize it; CTLS provides better retrieval quality at every query at no additional cost.

Several restrictions apply. The controlled evaluation uses a frozen single-layer bank; the production rail has different capacity constraints. The judge substitution and stratified subset are disclosed deviations from the official evaluation protocol. The blinded human-adjudication task remains incomplete and results use the primary predeclared judge. CTLS is established theoretically with empirical support from the `hybrid_no_time` ablation; a direct CTLS implementation and controlled evaluation remains the primary recommended follow-up. The present artifact records candidate scores and skipped-item behavior and can directly support that experiment.

7 CONCLUSION

We have identified temporal dominance as a structural failure mode in the deployed hybrid memory selector, derived the exact algebraic condition under which it occurs, and proposed CTLS as a principled correction. The deployed hybrid is outperformed by BM25 by -36.1111 percentage points, -29.3651 percentage points, and -33.7302 percentage points at the three evaluated token budgets in a controlled evaluation where only the selector changes. The temporal term causes most of this deficit: removing it (`hybrid_no_time`) recovers the majority of the gap. CTLS eliminates temporal dominance by construction through tier partition, preserving temporal decay as a within-tier tiebreaker.

The Non-Dominance Property proved in the method section guarantees that no zero-overlap item can outrank a positive-overlap item under CTLS, regardless of freshness gap, frequency, or importance values. This is a structural guarantee, not a hyperparameter constraint: it holds for any positive value of the temporal weight within the relevant-tier scoring function. CTLS requires no model calls, no embedding model, no learned parameters, and no write-path changes. It is a zero-cost selector

substitution that can replace the deployed hybrid in any context that uses the same frozen-bank, whole-item-packing retrieval architecture.

The practical contribution of this paper is the combination of a formal diagnosis and a constructive fix. The diagnosis identifies the exact condition under which a common retrieval heuristic fails, shows why the failure is invisible when evaluated only against a weak baseline, and shows that a large margin against a strong lexical baseline plus a clean ablation together pinpoint the cause. The fix is provably correct, immediately actionable, and requires no new infrastructure. Future work should directly evaluate CTLS on the existing frozen bank under the same controlled conditions, complete the blinded judge adjudication, and extend the mechanism analysis to multi-session and temporal-reasoning question types.

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A QUESTION-LEVEL AUDIT AND REPRODUCIBILITY

The following identifiers link directly to the submitted raw records. A and E denote answer correctness and complete-evidence recall. These are recorded single-run outcomes, not independent replications across budgets.

Table 6: Question audit at the 512-token budget.

Row	Original question ID	File A	File E	L1 A	L1 E
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Table 7: Question audit at the 1024-token budget.

Row	Original question ID	File A	File E	L1 A	L1 E
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Table 8: Question audit at the 2048-token budget.

Row	Original question ID	File A	File E	L1 A	L1 E
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Table 9: Question audit at the 512-token budget.

Row	Original question ID	File A	File E	L1 A	L1 E
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B OBSERVED DIAGNOSTIC CASES

Cases below are selected deterministically for diagnostic categories, not as a random qualitative sample. Answers are verbatim recorded outputs; questions are copied from the adapted dataset.

C ARTIFACT AND GENERATION PROVENANCE

The submission includes the adapted data, original write/query records, normalization code, citation metadata and a hashed source snapshot. Writing assistance uses a separate model from the experimental query model. Its actual requests, generated text and usage are saved separately; none of those calls contributes to the experimental accuracy. The methodology schematic is AI-generated and checked against the implementation; All empirical plots and tables are rendered programmatically from the recorded data.